ELVIS ALEXANDER AGÜERO VERA

MSc. in Applied Mathematics

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SCIENTIFIC WORK

1. *Impact of a rigid sphere onto an elastic membrane. Proceedings of the Royal Society A. https://doi.org/10.1098/rspa.2022.0340*

WORK EXPERIENCE

Visiting Research Student

Harris Lab @ Brown University

Jan-Feb 2023 — Providence, Rhode Island

- Executed fluid dynamics experiments, focusing on deformable droplet impact onto solid surfaces. Developed MATLAB simulation code to validate experimental outcomes. This research is a follow up of our previous published results on impacting phenomena.

Academic Consultant

Organización Multidisciplinaria de Apoyo a Profesores y Alumnos

Dec 2016 – Ongoing — Asunción, Paraguay

- Wrote more than 200 math problems for the Paraguayan National Math Olympiad, as well as International Math Competitions.
- Trainer and content writer for the "Young Talents Program", where the best students from the National Olympiad receive mathematical and scientific training for international competitions. Reference: Gabriela Gomez Pasquali.
- OMAPA is an non-profit NGO which promotes scientific education for high school students through competitions all around Paraguay. Its mathematical olympiad is taken by more than fifty thousand high school students from all over Paraguay.

Academic Project Manager

Piensa Paraguay

Dec 2020 – Ongoing — Ciudad del Este, Paraguay

- Designed the math curriculum and materials, mantaining a database with more than 1000 math problems, divided by difficulty, area and level. Reference: Osmar Quiñonez.

Data Analyst Consultant

NGO ReAcción

2020-2021 — Ciudad del Este, Paraguay

- ReAcción is a non-profit NGO that uses civic education and grassroots mobilization to demand accountability from government in the education sector.
- Used Python and Apps Script to analyze thousand of public contracts, comparing public expenditure of central and local institutions before and after the covid-19 outbreak. A couple of articles were made based on our findings. For more info, check out the following link (in spanish): `https://reaccion.org.py/publicaciones/#investigaciones`

EDUCATION

B.Sc., Engineering Physics

Universidade Federal de Integração Latinoamericana

2020 – Ongoing — Foz de Iguaçu, Brazil

M.Sc., Applied Maths

Instituto Nacional de Matematica Pura e Aplicada

2017 – Mar 2019 — Rio de Janeiro, Brazil

- Academic advisor: Andre Nachbin
- All internal IMPA requirements are completed to date. However, Brazil's goverment requires obtention of an undergraduate degree to grant the diploma

SKILLS AND PRODUCTIVITY

Python, Matlab, Julia, Scilab
Scientific computing, OOP, multi-purpose scripting (see my github for more info)

C++, Java, JS, Google Apps Script
Basic flow control and OOP

SQL, MongoDB, Access, Excel, Google Sheets
Basic data storage management

Git, Github
Professional Version control

LaTeX, Office packages
For typesetting high quality documents

TALKS AND TEACHING

- (2023) Contributed talk at the Brazil-China Joint Mathematical Meeting
- (2021) Teaching assistant for Numerical Calculus at UNILA.
- (2021) Speaker at the Paraguayan Mathematical Congress.
- (2021) Speaker at the series of talks "Hugo Checo, creando redes". **Best speaker award**.
- (2019) Teaching assistant of Functional analysis in the summer school at IMPA.

VOLUNTEER WORK AND OUTREACH

Math Olympiads Organizer, Jury member, Problem Author, Team Leader (2017 - Ongoing)

- Part of the Paraguayan delegation as professor in various International Olympiads, such as the Cyberspace Mathematical Competition (2020), IberoAmerican Olympiad (2021), and most recently at the International Mathematical Olympiad (2022), which was held in Oslo, Norway.
- Part of the jury committee at the XXXII Cono Sur Olympiad (2021), organized by Paraguay. Graded problems of all countries, and created problems for the contest.

Teaching, Board member: Mathura (2019 - 2022)

- Board member of Mathura, an NGO which gives extracurricular education to high school students in their pursuit for scholarships to attend undergraduate programs. Previously, part of the teaching staff of Mathura.

Junior Data Analyst: AIGA (2017 - 2021)

- AIGA, as part of ReAcción, emphasizes its work on Open Government Enhancement Strategies. Gave sameinars to the public to raise awareness on the importance of open data, and created a script to automate the process of contract data analysis.

AWARDS AND DISTINCTIONS

International Math Olympiad (IMO)

Bronze Medal

Jul 2016 — Hong Kong, Hong Kong

Bronze Medal

Jul 2015 — Chiang Mai, Thailand

IberoAmerican Math Olympiad
Bronze Medal (2016) and Honorific Mention (2015)

Cono Sur Olympiad
Bronze Medal (2015) and Honorific Mention (2014)

Paraguayan Math Olympiad
Three gold medals and four silver medals ($2010 - 2016$)

Prize "Domingo Martinez de Irala"
Order of Merit Medal, Asunción, 2017

Distinguished Citizen of Ciudad del Este Medal
2016

LANGUAGES

Spanish	CEFR C2
English	CEFR C1
French	CEFR A1
Guarani	Profficient
Portuguese	Profficient

FURTHER ACTIVITIES

- My account is percentile 0.5% in Project Euler, a problem-solving programming site.
- Attended three academic years of piano courses at “Conservatorio Mangoré”
- Course “Introduction to Philosophy”, completed in 2020, via “Coursera”
- Course “Algorithms, Part 1”, completed in 2020, via “Coursera”
- Course “Mastering Programming with MATLAB”, completed in 2020, via “Coursera”

REFERENCES

Prof. André Nachbin
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Eng. Osmar Quiñónez
✔ Piensa CEO
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Joel Prieto
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